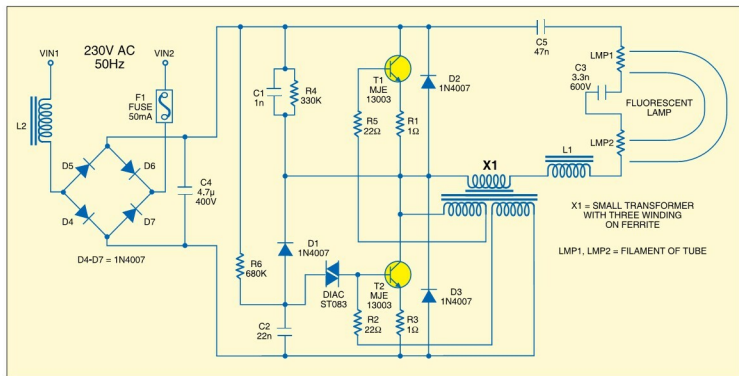




Typical starter circuit of a CFL

sustained in spite of the low voltage and the bulb continues to glow.

Q3. How does a satellite radio work? Is it the same as AM/FM radios which are commercially available in the market?

U. Sree Hari
Hyderabad

A3. Satellite radio uses a home-based ground broadcasting station that is uplinked to the satellites in an orbit above the Earth. These satellites transmit the radio signal, which is received by a portable satellite radio antenna and fed to the satellite radio receiver. The receiver decodes the received signal and outputs digital sound, along with information about the programme you are listening to, such as the song title and artist name.

Unlike AM/FM radio signals, satellite radio signals do not fade out with the distance. Also, with satellite radio you can listen to more than 100 channels of digital audio, including a multitude of musical genres and news services, many of which are commercial-free. Because it's digital audio, you can enjoy high-quality sound from your speakers.

You need to have the proprietary hardware to receive the satellite radio signal. Also, you have to subscribe to one of the service providers. Currently, there are three companies in the world that sell satellite radio subscriptions: WorldSpace Radio, XM Radio and Sirius Radio. In India, WorldSpace is the only company that provides satellite radio service.

Each WorldSpace satellite transmits three signal beams, carrying more than 60 channels of programming, to three overlapping coverage areas of about 14 million sq. km each. Each of the three beams can deliver crystal-clear audio and multimedia programming via the 1467-1492MHz segment of the L-Band spectrum which is allocated for digital audio broadcasting.

WorldSpace broadcasters uplink their signal to the operational satellite through a centralised hub site or an individual feeder link station located within the global uplink beam. The satellite then transmits the signal in one, two or all three beams. Receivers on the ground then pick up the signal and provide CD-quality sound through a detachable antenna.

WorldSpace satellite receivers are

capable of receiving data at a rate of 128 kbps. These use the proprietary StarMan chipset, manufactured by STMicroelectronics, to receive digital signals from the satellites.

You can contact the WorldSpace offices at the numbers given below to avail their subscriber services:

Corporate Office, Bangalore: Phone 080-22020000, Fax 22105030/22123483

North Regional Office, New Delhi: Phone 011-41608431-33, Fax 41721390

West Regional Office, Mumbai: Phone 022-26833252, 26833255/26849849/26822406/26823459, Fax 26848016

East Regional Office, Kolkata: Phone 033-22895922-24, Fax 22895920

South Regional Office, Bangalore: Phone 080-25548800/02/03, Fax 25548801

Since you are living in Hyderabad, you can contact the WorldSpace office in Hyderabad at 040-55622121.

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